

Extremely Large Gate Modulation in Vertical Graphene/WSe₂ Heterojunction Barristor Based on a Novel Transport Mechanism

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Despite the great interest in graphene with its superior electrical^[1] and mechanical properties,^[2] its semi-metallic nature (zero bandgap) that results in an extremely low on/off-current ratio has so far hindered the successful integration of graphene-based devices into mainstream electronics industry. Several approaches to widen the bandgap of graphene^[3–10] were proposed, but they usually resulted in the degradation of graphene mobility and the challenge of achieving a sufficiently high on/off-current ratio for circuit applications remains unresolved. In this light, another class of atomically thin 2D materials named transition metal dichalcogenides (TMDs) was proposed as an alternative. Having wider bandgaps than graphene, TMD materials like molybdenum disulfide (MoS₂) and tungsten diselenide (WSe₂) have been investigated as an alternative

ingredient for 2D nanoelectronics,^[11–30] demonstrating a high on/off-current ratio at room temperature.^[11] Recently, with high scalability down to tenth of nanometers, excellent junction interface quality, and low contact resistance, a new approach to 2D vertical heterojunction devices consisting of both graphene and TMD has been reported. This approach showed high potential in applications such as transistors,^[21–26] photovoltaic cells,^[27,28] and memory devices.^[29,30] In particular, for the vertical graphene-TMD heterojunction (GTH) transistors, the “barristor”^[21–25,27,28] and tunneling device^[26] architectures were proposed, whose characteristics were, respectively, understood based on the simple thermionic emission and direct tunneling models. However, most works related with GTH remained on the level of qualitative analysis of device operations, consequently hindering further optimization and development. In addition, all the GTH devices reported till present showed a reducing trend in on- and off-currents with temperature decrease and accordingly a modest low-temperature on/off-current ratio (below 10⁶), which was qualitatively explained by the suppression of thermionic emission at low temperatures.

Here, we report our discovery on the trap-assisted tunneling (TAT) process in a WSe₂-based vertical GTH barristor, which was optimized by suppressing charge puddles on graphene and controlling the thickness of WSe₂ multilayers. As a result of this TAT process, our WSe₂ GTH barristor shows an anomalously increasing on-current with decreasing temperature and achieves an extraordinarily large on/off-current ratio of 5×10^7 at 180 K (3×10^4 at room temperature). Based on first-principle density functional theory (DFT) calculations coupled with a temperature-dependent carrier density measurement and a newly proposed analytic device model, we for the first time quantitatively explain the atomistic origins of the low-temperature characteristics of the WSe₂ GTH barristor as well as the bias- and temperature-dependent device operations. We identify the dominant TMD trap states near the graphene Dirac point and attribute them to tungsten (W) vacancy defects. Furthermore, comparison with MoS₂-based GTH counterpart shows that the close match between graphene and WSe₂ work functions and the minimal effects of defects on the latter allow the trap states to dominate device characteristics. Our work demonstrates the critical role of defect engineering in TMD-based 2D devices and lays the foundation of understanding and optimizing various heterojunction 2D devices for next-generation electronics applications.

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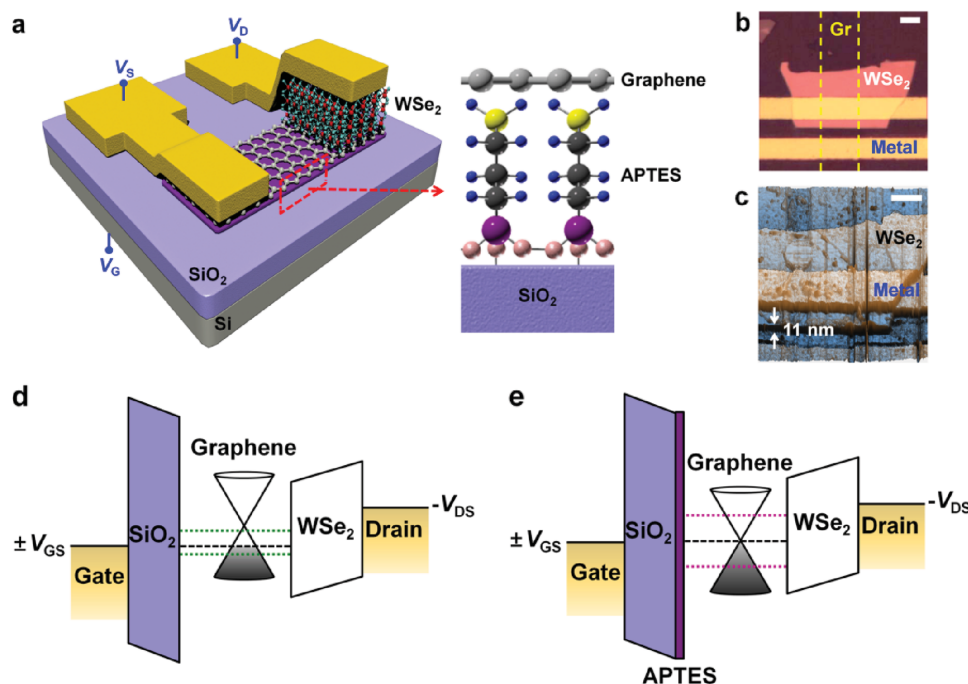


Figure 1. a) An illustration of WSe₂-based GTH barristor and a cross-sectional view of graphene/APTES layer/SiO₂ structure. Carbon, hydrogen, silicon, nitrogen, and oxygen are represented by gray, blue, purple, yellow, and pink spheres, respectively. b) Optical image and c) 3D atomic force microscope image of the GTH device. Scale bars are 5 μm. d) Band structure of GTH barristors with control graphene layer under negative V_{DS}. e) Band structure of GTH barristors with APTES-treated graphene layer under negative V_{DS}. The black dashed line indicates the Fermi-level of graphene under an equilibrium state (at V_{GS} = 0 V) and the green/magenta dashed lines mean Fermi-level shift by variation of V_{GS} between 20 and −20 V.

Figure 1a presents the schematic diagrams of our vertical GTH barristor. As a key preliminary step to achieve the novel device properties reported in this work, we fabricated a WSe₂-based vertical GTH device on the 3-aminopropyltriethoxysilane (APTES)-treated SiO₂ substrate. About 11-nm-thick WSe₂ layer was transferred onto the patterned graphene by a mechanical transfer process method^[31] and the metal electrodes were formed to make contact with the graphene and WSe₂ layers, as shown in the optical (Figure 1b and Supporting Information Figure S1) and 3D atomic force microscope (AFM) images (Figure 1c). Here, platinum (Pt), which has a very high work function, was used as the drain electrode to form a low hole barrier. Because a low hole barrier height (below 0.3 eV) is generally known to be formed at a WSe₂/Pt junction (i.e., an electron barrier height above 0.8–0.9 eV), the injection of electron carriers from the Pt electrode into WSe₂ is not expected to significantly affect the drain current.^[32] Device measurements, coupled with Raman spectroscopy and DFT calculations, were carried out to confirm the effect of the APTES layer on graphene (Supporting Information Figures S2–S4). Here, the electrical characteristics of the APTES-treated GTH barristor are expected to be improved because (i) the charge puddle effect is suppressed and (ii) graphene is lightly doped to n-type^[33] by inserting APTES layer between graphene and SiO₂, eventually recovering the properties of graphene to intrinsic (Supporting Information Figure S2). The intrinsic graphene has a low density of states (DOS) near the Fermi-level causing the larger amount of the Fermi-level shift (ΔE_F) by gate-biasing between −20 V and +20 V ($\Delta E_F = 0.32$ eV at V_{Dirac} = 0 V) than

that in the strongly p-doped graphene (0.11 eV at V_{Dirac} = 40 V), which is graphically described in Figure 1d,e. Fermi-level shift values were estimated by calculating the change in graphene charge density induced by V_{GS}, which is discussed in more detail in the Supporting Information. Since the Fermi-level shift of graphene directly affects the change of hole carrier density in graphene, the improved electrical characteristics in the APTES-treated GTH barristor (a tenfold increase in on-current, a 25-fold decrease in off-current, and consequently a 230-fold increase in on/off-current ratio in Supporting Information Figure S4) were achieved.

The drain current density–drain voltage (J_D –V_D) characteristics of the vertical GTH barristors, which were measured at 297 K, are shown in Figure 2a. Here, degradation of the electrical performance in the WSe₂ barristor was not observed, even after exposure to air for three months; WSe₂ is known to be stable in air at room temperature.^[34] Due to the low valence band edge of WSe₂ ($E_{VBM} = 5.2$ eV),^[35] a relatively low hole barrier is formed, which means that the source-to-drain current is dominated by hole transport. Under a negative gate voltage (V_{GS}), the Fermi level (E_F) of graphene is downshifted and the J_D is increased (on-state). Conversely, a positive V_{GS} up-shifts the graphene E_F and lowers the J_D (off-state). Figure 2b shows the low-temperature J_D –V_G characteristics of the APTES-treated GTH barristor (between 340 and 160 K), and the inset shows the J_D –V_G curves on a linear scale. Here, we observe a hitherto unreported phenomenon where the on-current density increases from 6 to 22 A cm^{−2} (V_{GS} = −20 V and V_{DS} = −0.1 V) as the measurement temperature decreases (Figure 2c), while

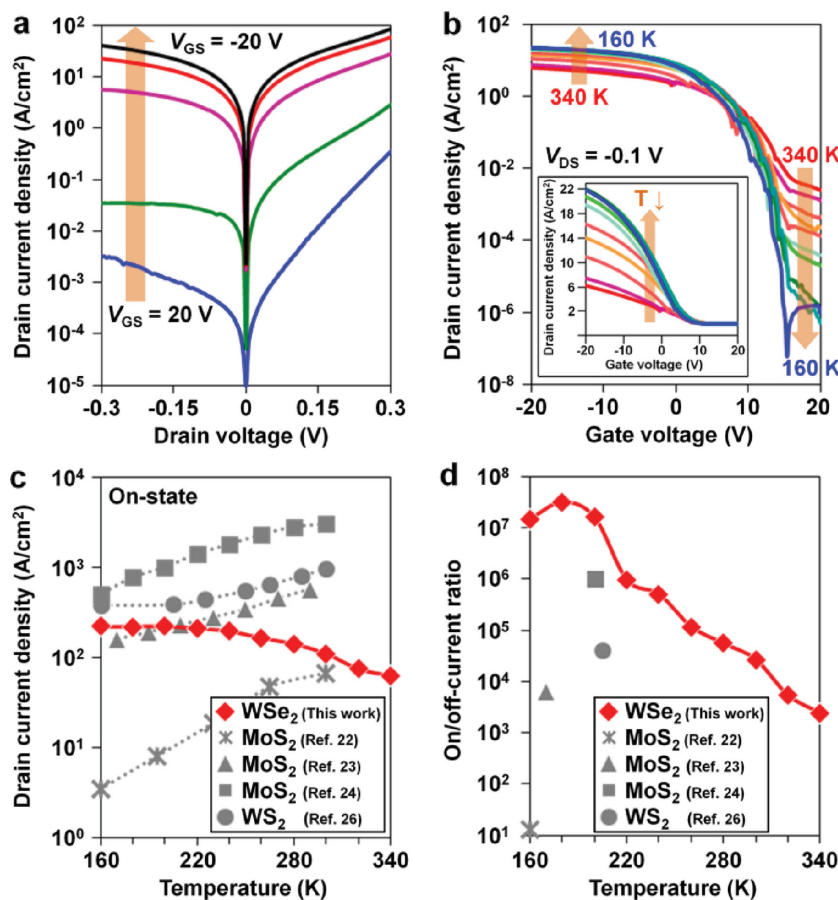


Figure 2. a) Electrical characteristics of the GTH barristor as a function of V_{DS} under various V_{GS} biases between -20 and 20 V. b) J_D - V_G curves of the GTH barristor at different temperatures between 340 and 160 K, where $V_{DS} = -0.1$ V. The inset shows the J_D - V_G curves on a linear scale. c) Measured drain current density of GTH barristor (red diamond symbols) in this work as a function of temperature, compared to extrapolated drain current densities (gray symbols) from GTH barristors of other groups,^[22–24,26] where the devices are in on-state. d) Extracted on/off-current ratio (red diamond symbols) of the GTH barristor in this work as a function of temperature; these values are compared to the extrapolated on/off-current ratios from other groups' data below 200 K (gray symbols).^[22–24,26]

the off-current density decreases. For reference, similar electrical characteristics were observed in three different WSe_2 barristor devices without or with an APTES layer. Remarkably, a WSe_2 barristor that was exposed to high (340 K) and low (160 K) measurement temperatures continued to show stable electrical characteristics when it was measured again at room temperature. In previous GTH barristors that were fabricated on MoS_2 , SnS_2 , etc., both the on- and off-state currents were simultaneously reduced as the temperature was decreased; this phenomenon has been predominantly explained by the suppression of thermionic emission.^[22–26] As a result, the on/off-current ratio values in these devices have been limited to below 10^6 , even when the temperature was decreased to 200 K. However, due to the unexpected increase in the on-current with a decrease in temperature (which cannot be explained by thermionic emission), an extremely high on/off-current ratio of 5×10^7 was observed in our WSe_2 -based GTH barristor below 180 K (Figure 2d). New qualitative and quantitative analyses are required to explain our findings.

To investigate the graphene/ WSe_2 GTH interface in detail, we performed temperature-dependent electrical measurements between 410 and 220 K in a WSe_2 -based lateral field-effect transistor (FET) device with a graphene source electrode and a long channel length (about $10 \mu\text{m}$). This was done to minimize the influence of the barrier-lowering effect on the drain current (inset of Figure 3a). Figure 3a shows the drain currents (I_D) as a function of $1/k_B T$ that was measured in the WSe_2 transistor (where the V_{DS} is -3 V and the V_{GS} is 0 V). In the I_D versus $1/k_B T$ graph, two different slopes (blue and green dashed lines) were observed, indicating the presence of two or more traps in WSe_2 .^[20] The number of trap levels in WSe_2 is related to the number of slopes shown in the drain current (I_D) of the FET versus $1/k_B T$ graph. In addition, the degree of the slope indicates the amount of energy that is required to ionize traps. Here, we note that W or selenium (Se) vacancies are known to behave as acceptor-like traps.^[36,37] If acceptor-like traps with various energy levels exist in WSe_2 , the densities of each ionized trap should be varied as a function of temperature, consequently affecting the drain current. Based on an iterative calculation using the Fermi functions and the densities of traps, which are explained in detail in Supporting Information Figures S5 and S6, we estimated the values for the trap densities and corresponding energy levels in WSe_2 as $1.05 \times 10^{12} \text{ cm}^{-2}$ at $E_{\text{VBM}} + 0.53$ eV (deep acceptor level) and $1.08 \times 10^7 \text{ cm}^{-2}$ at $E_{\text{VBM}} + 0.09$ eV (shallow acceptor level). As the temperature is reduced, I_D decreases because the ionization of acceptor-like trap is reduced, resulting in the decrease in internal electric field between graphene and WSe_2

and consequently the increase in contact resistance (Supporting Information Figure S7). Especially, below 280 K, I_D decreases with a relatively gentle slope (green dashed line). While the number of ionized deep traps decreases as the temperature is reduced, most of the shallow traps of WSe_2 which are located below E_F should be ionized in the entire temperature range. As a result, total ionized trap density approaches to shallow trap density as temperature reduces (Supporting Information Figure S8), subsequently leading to the relatively gentle decrease in I_D below 280 K (above $1/k_B T = 41$).

The atomistic nature of the graphene/ WSe_2 GTH interface was further analyzed based on DFT calculations. To achieve quantitative agreements with the experimentally identified defect level positions, we employed the self-interaction-corrected local density approximation (LDA) scheme (Supporting Information).^[38] Three WSe_2 layers were used to properly reproduce experiments based on multilayer WSe_2 , and the Se and W vacancies introduced in the first WSe_2 interfacial layer were considered as defect candidates (Supporting

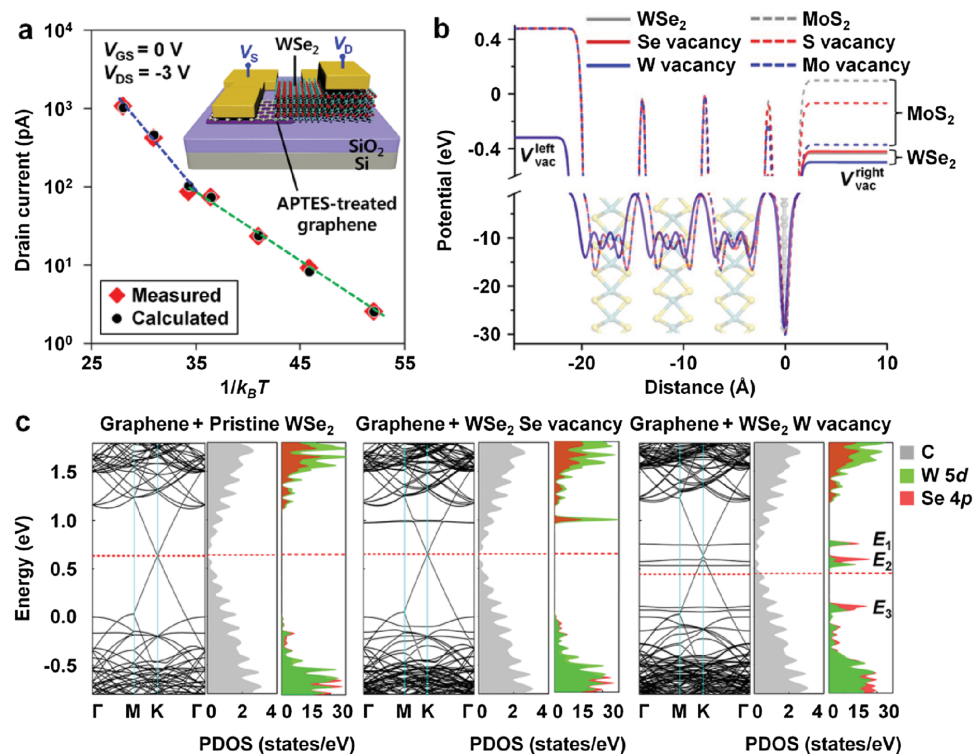


Figure 3. a) Measured (red diamond symbols) and calculated (black circle symbols) I_D versus $1/k_B T$ curve for WSe₂-based lateral FET with a graphene source electrode, where $V_{GS} = 0$ V and $V_{DS} = -3$ V. The blue and green dashed lines indicate two different slopes. The inset shows the structure of the WSe₂ lateral device. b) DFT-derived plane-averaged electrostatic potentials of the GTH interfaces based on WSe₂ and MoS₂ in different TMD conditions: pristine WSe₂ (gray solid line), WSe₂ with a Se vacancy (red solid line), WSe₂ with a W vacancy (blue solid line), pristine MoS₂ (gray dashed line), MoS₂ with an S vacancy (red dashed line), and MoS₂ with a Mo vacancy (blue dashed line). c) Band structure, graphene projected DOS (PDOS), and the first interfacial-layer WSe₂ PDOS of the GTHs based on pristine and defected (with a Se or W vacancy) WSe₂. The red dotted lines represent graphene E_F in each case. In the WSe₂ PDOS plots, we show the PDOS of W 5d orbitals (green filled lines) and the Se 4p orbitals (red filled lines), separately.

Information Figure S10). We present the potential profiles of the three GTH interface models based on pristine and defected WSe₂ in Figure 3b, and the associated band structure and DOS are projected onto graphene and the first WSe₂ interfacial layer in Figure 3c. Figure 3c shows that the trap states identified in the experiment originate from the W vacancy defect. Specifically, we find in the W vacancy GTH model three defect states at $E_1 = E_{VBM} + 0.76$ eV, $E_2 = E_{VBM} + 0.53$ – 0.57 eV, and $E_3 = E_{VBM} + 0.07$ – 0.11 eV, of which the latter two are in excellent quantitative agreement with experimental values. The E_2 and E_3 defect states, which respectively comprise two bands associated with the W and Se atoms neighboring the W vacancy, are delocalized across the WSe₂ layer (Supporting Information Figure S12), and should be a very good medium for charge tunneling across it.

In Figure 3b, the potential value on the right vacuum region represents the work function of graphene in contact with multilayer WSe₂. Notably, we find only a small difference between graphene and WSe₂ work functions as well as within the WSe₂ defect type-dependent graphene work functions. Accordingly, in the band structure plots of Figure 3c, the corresponding graphene E_F as well as the graphene Dirac point energy (E_D) or the alignment of graphene Dirac cone with respect to WSe₂ bands also shows small variations regardless of the WSe₂ models. This is in stark contrast to the behavior of the MoS₂-based GTH interface, which shows a larger difference

between graphene and pristine MoS₂ work functions and very large variations in the MoS₂ work functions and associated E_D and E_F values with the introduction of S and Mo vacancies (Figure 3b; see also Supporting Information Figure S13). The graphene/Mo-vacancy MoS₂ model in fact shows defect levels that are similar to those from the graphene/W-vacancy WSe₂ counterpart. However, in the case of graphene/MoS₂ interface, the Mo vacancy defect states should be irrelevant for the majority of electrons energetically located near E_F of the interfaces based on pristine MoS₂ or MoS₂ with S vacancies, which are more dominant defects than Mo vacancies.^[39] Such significant variations in the relative GTH electronic structure with respect to MoS₂ defect types should destroy any coherence in the energy level alignment throughout the graphene/MoS₂ 2D interface and preclude any device operation principle of quantum origin such as tunneling. Apparently, this is not the case for the GTH interface based on WSe₂.

Based on the atomistic information obtained in DFT calculations, we now explain the novel temperature- and bias-dependent device characteristics of our GTH barristor by developing a theoretical analysis model. Figure 4a shows the energy band diagram of the GTH barristor under positive V_{GS} and negative V_{DS} . Here, because a thin WSe₂ layer (11-nm-thick) was used in the GTH barristor, the WSe₂ channel is predicted to be fully depleted. There are three possible carrier

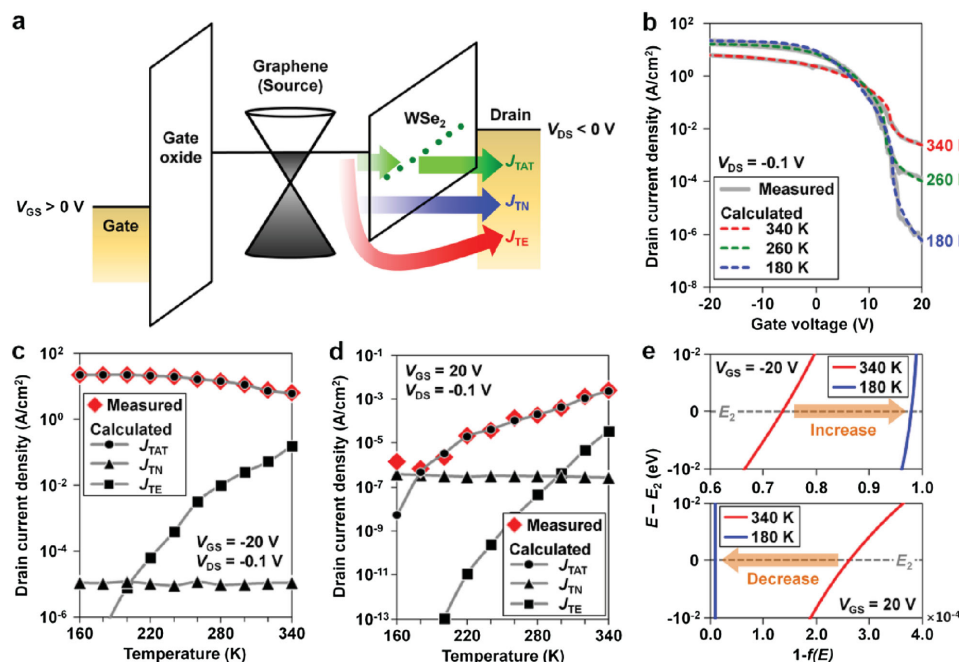


Figure 4. a) Energy band diagram of GTH barristor under positive V_{GS} and negative V_{DS} biases. Green circles indicate deep acceptor-like traps. Green, blue, and red arrows indicate J_{TAT} , J_{TN} , and J_{TE} , respectively. b) Experimentally measured (gray solid lines) and theoretically calculated J_D - V_G curves of the GTH barristor at 340 K (red dashed line), 260 K (green dashed line), and 180 K (blue dashed line), where $V_{DS} = -0.1$ V. c,d) Measured drain current density (red diamond symbols) and calculated J_{TAT} (black circle symbols), J_{TN} (black triangle symbols), and J_{TE} (black square symbols) of the GTH device at c) $V_{GS} = (20$ V and d) $V_{GS} = 20$ V, where $V_{DS} = -0.1$ V. e) The probability that a state is empty as a function of a given energy E relative to E_2 ($E - E_2$) under negative V_{GS} (top) and positive V_{GS} (bottom) at 340 K (red solid line) and 180 K (blue solid line).

transport mechanisms (from graphene to WSe_2), which are (i) trap-assisted tunneling current density (J_{TAT}), (ii) Fowler–Nordheim (F–N) & direct tunneling current density (J_{TN}), and (iii) thermionic emission current density (J_{TE}). The J_{TAT} was calculated using the densities of traps and the corresponding trap levels (E_1 , E_2 , and E_3) that were obtained from the above-discussed temperature-dependent electrical measurements and DFT calculations (Figure 3a,c). The J_{TN} and J_{TE} are found by calculating the barrier height between graphene and WSe_2 (Φ_B), which is expressed as a function of V_{GS} because the Fermi level of graphene is controlled by adjusting V_{GS} . The equations related to the current transport mechanisms can be found in the Supporting Information. The measured temperature-dependent current behavior is well matched with the J_D - V_G curves estimated by the analytic model (at 340, 260, and 180 K), as shown in Figure 4b. In order to understand the operation mechanism of the WSe_2 -based GTH barristors in more detail, we investigated each current component (J_{TAT} , J_{TN} , and J_{TE}) in on- (Figure 4c) and off-states (Figure 4d). We conclude that, under both negative and positive V_{GS} , J_D in the GTH barristor mainly consists of J_{TAT} over the entire measurement temperature range (between 340 and 160 K). It is because J_{TN} and J_{TE} will be suppressed due to the relatively large barrier height of 0.51 eV from graphene to WSe_2 (assuming the pinning of graphene E_F near E_2 ; Figure 3c). Here, because the trap density of WSe_2 in E_2 ($1.05 \times 10^{12} \text{ cm}^{-2}$) is higher than the densities of other traps in E_1 and E_3 (10^3 cm^{-2} in E_1 and $1.08 \times 10^7 \text{ cm}^{-2}$ in E_3), the tunneling current through the deep traps in E_2 level is thought to be most dominant. The

existence of traps in E_1 was not identified through the temperature-dependent analysis (Figure 3a) because of their low concentration. When a negative V_{GS} is applied, the graphene E_F moves downward and it increases the probability of states near the E_2 level being empty (occupied by hole carriers). This consequently increases J_{TAT} , which contributes to the on-state of the barristor (Supporting Information Figure S15a). On the contrary, under positive V_{GS} that drives the barristor into the off-state, the graphene E_F is lifted upward and the probability of states near the E_2 level being empty decreases, resulting in a reduction in J_{TAT} (Supporting Information Figure S15b). Especially, as the temperature drops, the Fermi–Dirac distribution near E_F becomes sharp and thereby the probability of states near the E_2 level being empty increases (from 0.73 at 340 K to 0.97 at 180 K), finally causing higher J_{TAT} under a negative V_{GS} (top of Figure 4e). We also note that the increase in the contact resistance at low temperatures, which is explained in Supporting Information Figure S7 as being caused by the weakened image charging effect, can be ignored under a negative V_{GS} (on-state) because a very low barrier height (about 0.3 eV) is expected to be formed. Alternatively, under a positive V_{GS} , the J_{TAT} is reduced as the temperature decreases because the probability of the states near the E_2 being empty, which is now located below the graphene E_F , decreases at low temperatures (from 2.78×10^{-4} at 340 K to 1.45×10^{-8} at 180 K), as shown in the bottom of Figure 4e. In addition, the increased contact resistance caused by the suppressed image charging phenomenon (Supporting Information Figure S7) is expected to reduce the drain current as the temperature decreases. Overall, in the

WSe₂-based GTH barristor, the anomalous increase in the on-current and the decrease in the off-current with dropping the temperature can be explained by the temperature dependence of graphene's hole carrier distribution near the dominant WSe₂ defect energy levels. This novel operation mechanism differentiates our WSe₂-based GTH barristor from previous GTH barristors that were based on the modulation of barrier height via gate biasing.^[21–26] Previously reported GTH barristors appear to have relatively low barrier heights and/or low TMD trap densities; as a result, these devices operate via mechanisms of thermionic emission or direct/F–N tunneling. Moreover, even if trap densities are high, we explained above (Figure 3b,c and Supporting Information Figure S13, and associated texts) that large variations of local electronic structure depending on defect types should make TAT involving these states irrelevant, leaving thermionic emission or direct/F–N tunneling as device operation mechanisms.

We discovered a new charge transport mechanism (TAT) in the WSe₂-based vertical GTH barristor that was optimized by suppressing charge puddle effects and controlling the WSe₂ thickness. This consequently caused an anomalous low-temperature device characteristic, which showed an on-current increase (from 6 A cm^{−2} at 340 K to 22 A cm^{−2} at 160 K) with decreasing temperature, resulting in an unprecedented high on/off-current ratio of 5×10^7 at 180 K. This behavior is in stark contrast to previously reported GTH barristors employing other TMD materials that showed a reducing trend in both on- and off-currents with temperature decrease and accordingly a modest low-temperature on/off-current ratio through thermionic emission process.^[22–26] DFT calculations showed that the small difference between graphene work function and those of pristine and defected WSe₂ allows the defect states originating from W vacancies to dominate the electronic structure of the graphene–WSe₂ interfaces. Additional temperature-dependent carrier density measurements and theoretical analysis then confirmed that the TAT process is the main operating mechanism of the WSe₂-based GTH barristor. These observations and related analyses, which demonstrate that TMD bulk point defects can be actively employed to realize novel device functionalities, should be able to contribute in designing future electronic and optoelectronic applications based on the GTH devices.

Experimental Section

APTES Treatment on SiO₂/Si Substrate: 600 μL APTES was added to 25 mL toluene, and the SiO₂/Si substrates were soaked in the solution for 1 h. Then, the samples were rinsed and sonicated several times with toluene, alcohol, and deionized water, followed by a baking process at 120 °C for 20 min.

Fabrication of the WSe₂-Based GTH Barristors: The graphene was grown on copper (Cu) foil by a chemical vapor deposition process, and the bottom Cu foil was etched in ferric chloride (FeCl₃). The graphene was then transferred onto an APTES-treated 90 nm thick SiO₂/Si substrate. The transferred graphene was patterned using photolithography and oxygen plasma etching processes. Next, WSe₂ flake was transferred onto the patterned graphene through a mechanical transfer process. Finally, Pt/Au (10/20 nm) layers were deposited to form source and drain regions in an electron-beam evaporating system.

Characterization of the WSe₂-Based GTH Barristors: Raman studies were conducted using a WITec micro-Raman spectrometer system with a frequency-doubled Nd-YAG laser beam (532 nm laser excitation). AFM was carried out using an XE 100 (Park Systems Corp.) system. The electrical transport measurements were conducted with a probe station using a Hewlett-Packard (HP) 4155A Semiconductor Parameter Analyzer at room temperature. The temperature-dependent electrical characteristics were measured using a Keithley 4200 Semiconductor Parameter Analyzer. All theoretical calculations were conducted by using Matlab.

Density Functional Theory Calculations: DFT geometry optimizations were performed within the LDAs^[40] as well as generalized gradient approximation (GGA) supplemented by the Grimme DFT-D2 van der Waals correction scheme^[41] using the SIESTA software.^[42] Core electrons were removed by Troullier-Martins-type norm-conserving nonlocal pseudopotentials,^[43] and the Kohn–Sham pseudo-orbitals were expended in terms of double- ζ -plus-polarization-level numerical atomic basis sets defined by the confinement energy of 80 meV. The parameters for W pseudopotential were taken from ref. [44]. Real-space integration was performed with a mesh cutoff energy of 300 Ry. To calculate the electronic structure of graphene–WSe₂ and graphene–MoS₂ GTH interfaces, the LDA-based atomic self-interaction correction (ASIC) scheme^[27] was employed using the SMEAGOL code.^[45] In general, because of the self-interaction error, LDA and GGA underestimate band gaps of semiconductors and in their heterostructures could provide inaccurate band alignment. Particularly, the energetic position of defect levels and spatial spread of defect states can be incorrectly described in LDA and GGA. For example, the LDA or GGA underestimation of conduction band edge can spuriously place defect states inside the conduction band of host crystal.^[46,47] ASIC has been proved to remedy these deficiencies for various systems^[46,47] so LDA-ASIC calculations were performed to quantitatively analyze the band alignment and location of TMD-originated defect states in the graphene–TMD interfaces. For the ASIC scaling parameter α ($\alpha = 0$ and 1 correspond to LDA and full ASIC-LDA, respectively), $\alpha_{\text{Mo}} = \alpha_{\text{S}} = 0.5$, a value which is generally appropriate for mid-gap insulators, was previously suggested for MoS₂.^[48] However, for WSe₂, the α values of individual W and Se atoms were more systematically adjusted, and $\alpha_{\text{W}} = 0.5$ and $\alpha_{\text{Se}} = 0.1$ were chosen which closely reproduced the literature values of WSe₂ work function as well as bandgap.^[49] Choosing similar $\alpha_{\text{Mo}} = 0.5$ and $\alpha_{\text{S}} = 0.1$ for MoS₂, these parameters produced for the WSe₂ and MoS₂ monolayers the work functions of 4.58 and 5.39 eV, respectively, which are in good agreement with experiments.^[50,51] Calculated bandgaps of WSe₂ and MoS₂ monolayers were 1.84 and 1.98 eV, respectively, which are close to those obtained with the hybrid Heyd–Scuseria–Ernzerhof exchange correlation functional.^[52] Details of the procedure of choosing α , α -dependent computational results, and their comparison with experimental and literature data will be provided in a separate publication.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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